Sandith Ganhewage

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Education

Duke University Durham, NC
Bachelor's in Computer Science and Electrical & Computer Engineering *Expected Grad: May 2028*
- GPA: 3.98/4.00, Dean's List with Distinction (F24, S25, F25, S26), J. Welch Hariss Scholar
- Relevant Coursework: Data Structures & Algorithms, Advanced Computer Architecture, Linear Algebra, Deep Learning

Experience

Amazon Web Services (AWS) May 2026 – Aug. 2026
Software Development Engineer Intern *Raleigh, NC*
- Designed cache visibility tooling on an internal production debugging CLI for synchronization daemons on AWS network switches
- Implemented runtime inspection of object modules, exposing state during live production incidents without interrupting service
- Achieved low-overhead runtime diagnostics through efficient streams and lock minimization, preserving system performance

o11 Sep. 2025 – Mar. 2026
Founding Engineer *San Francisco, CA*
- Built full-stack AI platform for bankers, enabling ingestion, vector embeddings, and retrieval of proprietary financial documents
- Architected production RAG pipeline, implementing semantic chunking and embedding for scalable retrieval
- Developed Office 365 tools powered by RAG pipelines for document generation using firm-specific financial context
- Received backing from Y Combinator (W26), LeapYear, and KPMG, supporting enterprise pilots and product launch

narb Inc. Aug. 2025 – Sep. 2025
Software Engineering Intern *Remote*
- Designed user application for multi-modal LLM technology, utilizing Convex for a 30% increase in development speed
- Migrated existing applications to a new cross-platform framework prioritizing rapid development and implementation

Qorvo Inc. May 2025 – Aug. 2025
Test Engineering Intern *Greensboro, NC*
- Designed an embedded systems solution to chip sorting with the Synax S9 Handler, reducing configuration time by 25%
- Deployed company-wide solution for 50+ employees through an intranet-hosted web interface with React and Node.js
- Implemented low-level Arduino GPIB interface, alongside a Raspberry Pi connecting to universal web UI via bluetooth

Duke Center for Computational and Evolutionary Intelligence Jan. 2025 – May 2025
Undergraduate Researcher *Durham, NC*
- Tested and researched machine learning model quantization and distillation techniques to reduce model size by up to 30% and inference latency by up to 10% while maintaining model accuracy using PyTorch libraries
- Researched applications in 5 established bioinformatics algorithms for hardware acceleration by tri-state CAMs

Projects

Low-Latency Electronic Trading Engine | *C++, CMake, Sockets, Python, gtest*
- Built single-symbol limit order book in C++; profiling-driven optimizations, boosting throughput by 24% after local testing
- Replaced per-order heap allocation with a custom slab pool and lock-free producer-consumer ring with cache-aligned atomics, cutting worst-case latency by 41%
- Built TCP gateway with a fixed 40-byte binary message format shared between the network layer and internal queue, removing parsing overhead on the critical path
- Validated correctness with 20+ unit tests including a 5000-iteration randomized property test checking book state

Duke Automatic Code Grading Interface (ADTG) | *Spring Boot, Thymeleaf, PostgreSQL*
- Developed automatic delivery, testing, and grading system alongside Duke Professor for 100+ student graduate classes
- Implemented grading infrastructure for instructors/TAs, building reliable backend logic for secure database transactions
- Diagnosed and resolved UI and database issues across the application, improving system reliability and user experience

MeetUp | *React, React Native, Tailwind CSS, Convex, Clerk, Google/Outlook OAuth*
- Built cross-platform scheduling app with Google/Outlook calendar integration, simplifying event coordination
- Implemented dynamic group scheduling and unified profiles to accelerate onboarding and collaboration
- Secured user authentication with Clerk and designed privacy controls for configurable data sharing across groups

Leadership & Certifications

AWS Certified AI Practitioner (AIF-C01) – Amazon Web Services Aug. 2026
Duke Catalyst – Pre-professional Technology Society Jan. 2025 – Present
Duke Applied Machine Learning – Machine Learning & Systems Projects Sep. 2024 – Dec. 2025
Duke Law School – Instructional Technologies Consultant Aug. 2025 – Dec. 2025

Technical Skills

Languages: TypeScript, Python, Java, Rust, SQL, C/C++
Frameworks & Libraries: React, Next.js, Node.js, Spring Boot, PyTorch, shadcn/ui
Databases & Cloud: PostgreSQL, MongoDB, S3, Lambda, EC2, Azure, Docker
Systems & Infrastructure: CI/CD, Redis, BullMQ, RAG pipelines